

Pulse Recovery

Turning heart-rate recovery data into practical training guidance

Pulse Recovery helps users answer a simple but important question after exercise: *how well did I recover, and what should I do next?*

Many apps and devices can record heart-rate data, but the harder problem is helping users understand what that data means. Pulse Recovery closes that gap by combining smartwatch heart-rate recovery data, subjective feedback, session history, and recovery trends to provide clear, personalised guidance after exercise.

What it does

Pulse Recovery uses the watch as a low-friction capture device and the phone as the place where the user reviews and understands the result.

The watch records heart-rate data during exercise and recovery. Once the session is synced to the phone, the app shows the full workout and recovery curve, including peak heart rate, end-of-exercise heart rate, first-minute recovery, second-minute recovery, and the overall shape of the recovery curve.

The user can then add subjective context, such as how hard the session felt and how they felt afterwards. This matters because two workouts can produce similar heart-rate numbers but feel very different to the person doing them.

The app turns this combination of objective and subjective data into advice at two levels:

- **Session advice:** what the latest workout appears to suggest.
- **Trend advice:** how the user's recovery pattern appears to be changing over time.

What makes it different

Pulse Recovery is not just a heart-rate chart or a generic recovery score. It interprets the latest session, explains what appears to be driving the result, compares it with previous recovery patterns, and helps the user decide how to approach their next workout.

The advice is generated by an explainable rule-based decision engine rather than a black-box AI response. This makes the guidance transparent, consistent, testable, and tunable. AI may later be useful as a coaching-language layer, but the core recovery judgement should remain grounded in clear decision logic.

Current status

Pulse Recovery has progressed beyond a concept into a working product prototype, with:

- smartwatch-based heart-rate session capture
- phone-based session review
- full workout and recovery charting
- calculated recovery metrics

- perceived effort and post-exercise feeling inputs
- session-level advice
- trend-level advice
- session history
- Firebase-backed data persistence
- iOS / Apple Watch and Android / Galaxy Watch development

Future direction

Future development will focus on stronger personal baselines, richer trend analysis, improved chart export, clearer workout-type context, and more adaptive coaching language.

Pulse Recovery demonstrates end-to-end product thinking: identifying a real user problem, designing a watch-and-phone workflow, capturing physiological data, building an explainable decision engine, and turning raw measurements into practical user guidance.